

Assignment 3: Data Exploration

Flannery McNair

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
library(tidyverse)
library(here)
library(lubridate)

getwd()
```

```
## [1] "/Users/flannerymcnair/Documents/ENVIRON 872L/872 Class Documents"
```

```
Neonics <- read.csv(file = '/Users/flannerymcnair/Documents/ENVIRON 872L/872 Class Documents/Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv',
                    stringsAsFactors = TRUE)
```

```
Litter <- read.csv(file = '/Users/flannerymcnair/Documents/ENVIRON 872L/872 Class Documents/Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv',
                  stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: The effect of insecticides on insects can be relevant when looking at changes in biodiversity or when implementing ecological restoration. Additionally, the effect of insecticides on insects can have changes on the landscape or other organisms that is important to be studied.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: This can be helpful when studying wildfires and their causes because oftentimes increased woody debris can be the cause of more intense wildfires. Additionally, this can impact how plants grow and evolve in those areas.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Sampling only occurred in tower plots. 2. There were usually 1-4 litter traps per plot. 3. Ground traps were sampled once per year.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dimensions <- dim(Neonics)
dimensions
```

```
## [1] 4623 30
```

6. Using the `summary` function on the "Effect" column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
summary(Neonics$Effect)
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry
##           12           102           360           11
##      Cell(s)      Development      Enzyme(s) Feeding behavior
##           9           136           62           255
##      Genetics      Growth      Histology      Hormone(s)
##          82           38           5           1
##      Immunological      Intoxication      Morphology      Mortality
##          16           12           22           1493
##      Physiology      Population      Reproduction
##           7           1803           197
```

```
common_effects <- sort(summary(Neonics$Effect), decreasing = TRUE)
common_effects
```

```
##      Population      Mortality      Behavior Feeding behavior
##          1803          1493          360           255
```

##	Reproduction	Development	Avoidance	Genetics
##	197	136	102	82
##	Enzyme(s)	Growth	Morphology	Immunological
##	62	38	22	16
##	Accumulation	Intoxication	Biochemistry	Cell(s)
##	12	12	11	9
##	Physiology	Histology	Hormone(s)	
##	7	5	1	

Answer: Population and Mortality are the most common effects studied. This makes sense because insecticides would likely have large impacts on these two variables, due to their goal being to kill insects.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
Neonics$Species.Common.Name <- as.factor(Neonics$Species.Common.Name)
```

```
summary(Neonics$Species.Common.Name)
```

##	Honey Bee	Parasitic Wasp
##	667	285
##	Buff Tailed Bumblebee	Carniolan Honey Bee
##	183	152
##	Bumble Bee	Italian Honeybee
##	140	113
##	Japanese Beetle	Asian Lady Beetle
##	94	76
##	Euonymus Scale	Wireworm
##	75	69
##	European Dark Bee	Minute Pirate Bug
##	66	62
##	Asian Citrus Psyllid	Parastic Wasp
##	60	58
##	Colorado Potato Beetle	Parasitoid Wasp
##	57	51
##	Erythrina Gall Wasp	Beetle Order
##	49	47
##	Snout Beetle Family, Weevil	Sevenspotted Lady Beetle
##	47	46
##	True Bug Order	Buff-tailed Bumblebee
##	45	39
##	Aphid Family	Cabbage Looper
##	38	38
##	Sweetpotato Whitefly	Braconid Wasp
##	37	33
##	Cotton Aphid	Predatory Mite
##	33	33
##	Ladybird Beetle Family	Parasitoid
##	30	30
##	Scarab Beetle	Spring Tiphia
##	29	29
##	Thrip Order	Ground Beetle Family
##	29	27

##	Rove Beetle Family	Tobacco Aphid
##	27	27
##	Chalcid Wasp	Convergent Lady Beetle
##	25	25
##	Stingless Bee	Spider/Mite Class
##	25	24
##	Tobacco Flea Beetle	Citrus Leafminer
##	24	23
##	Ladybird Beetle	Mason Bee
##	23	22
##	Mosquito	Argentine Ant
##	22	21
##	Beetle	Flatheaded Appletree Borer
##	21	20
##	Horned Oak Gall Wasp	Leaf Beetle Family
##	20	20
##	Potato Leafhopper	Tooth-necked Fungus Beetle
##	20	20
##	Codling Moth	Black-spotted Lady Beetle
##	19	18
##	Calico Scale	Fairyfly Parasitoid
##	18	18
##	Lady Beetle	Minute Parasitic Wasps
##	18	18
##	Mirid Bug	Mulberry Pyralid
##	18	18
##	Silkworm	Vedalia Beetle
##	18	18
##	Araneoid Spider Order	Bee Order
##	17	17
##	Egg Parasitoid	Insect Class
##	17	17
##	Moth And Butterfly Order	Oystershell Scale Parasitoid
##	17	17
##	Hemlock Woolly Adelgid Lady Beetle	Hemlock Woolly Adelgid
##	16	16
##	Mite	Onion Thrip
##	16	16
##	Western Flower Thrips	Corn Earworm
##	15	14
##	Green Peach Aphid	House Fly
##	14	14
##	Ox Beetle	Red Scale Parasite
##	14	14
##	Spined Soldier Bug	Armoured Scale Family
##	14	13
##	Diamondback Moth	Eulophid Wasp
##	13	13
##	Monarch Butterfly	Predatory Bug
##	13	13
##	Yellow Fever Mosquito	Braconid Parasitoid
##	13	12
##	Common Thrip	Eastern Subterranean Termite
##	12	12

```
##           Jassid           Mite Order
##           12           12
##           Pea Aphid           Pond Wolf Spider
##           12           12
##           Spotless Ladybird Beetle           Glasshouse Potato Wasp
##           11           10
##           Lacewing           Southern House Mosquito
##           10           10
##           Two Spotted Lady Beetle           Ant Family
##           10           9
##           Apple Maggot           (Other)
##           9           670
```

```
top6_common_species <- head(sort(summary(Neonics$Species.Common.Name), decreasing = TRUE), 7)
top6_common_species
```

```
##           (Other)           Honey Bee           Parasitic Wasp
##           670           667           285
## Buff Tailed Bumblebee   Carniolan Honey Bee           Bumble Bee
##           183           152           140
##           Italian Honeybee
##           113
```

Answer: The top most common are honey bee, parasitic wasp, buff tailed bumblebee, carniolan honey bee, bumble bee, and italian honeybee. They are all types of bees. Bees are very important to ecosystem health because of their significant roles as pollinators.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

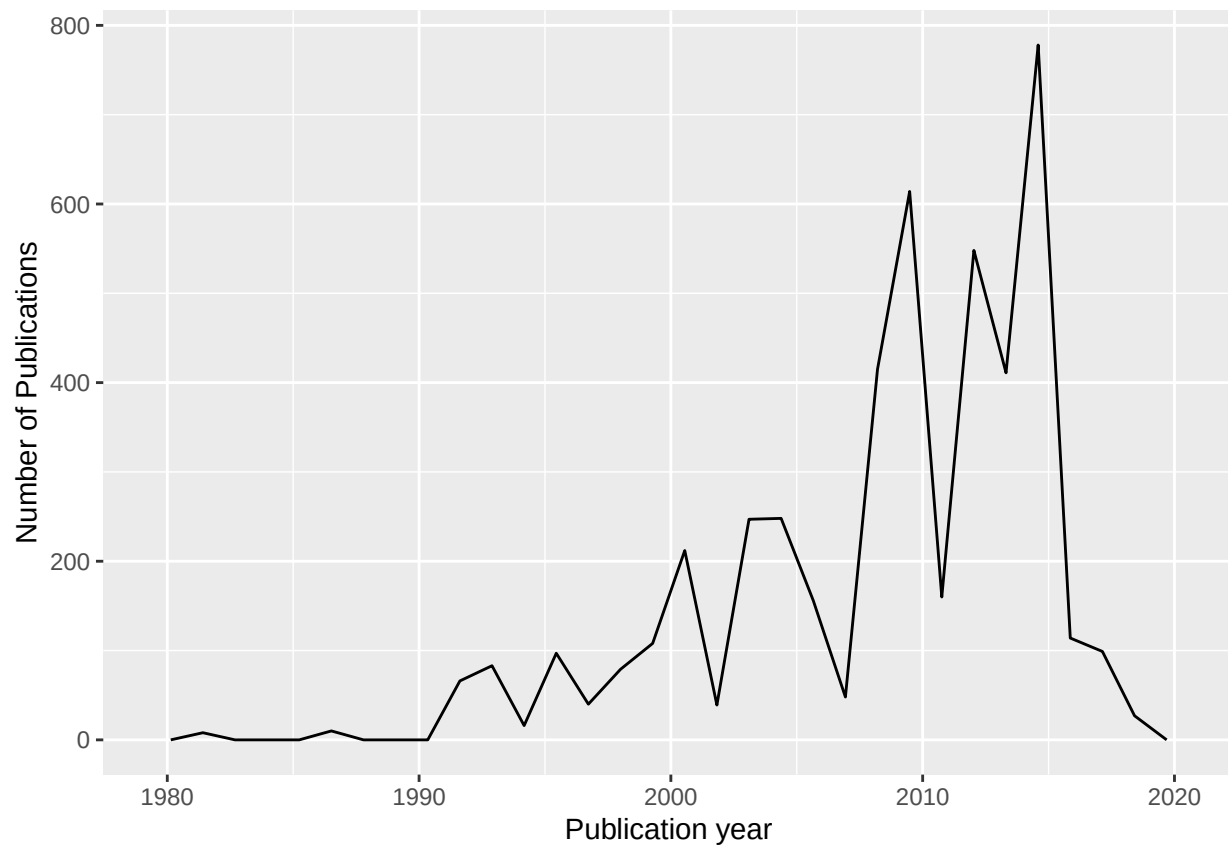
Answer: Factor, there are slashes in some of them so they can't be numeric.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(Neonics, aes(x=Publication.Year)) +
  geom_freqpoly() +
  xlab("Publication year") +
  ylab("Number of Publications")
```

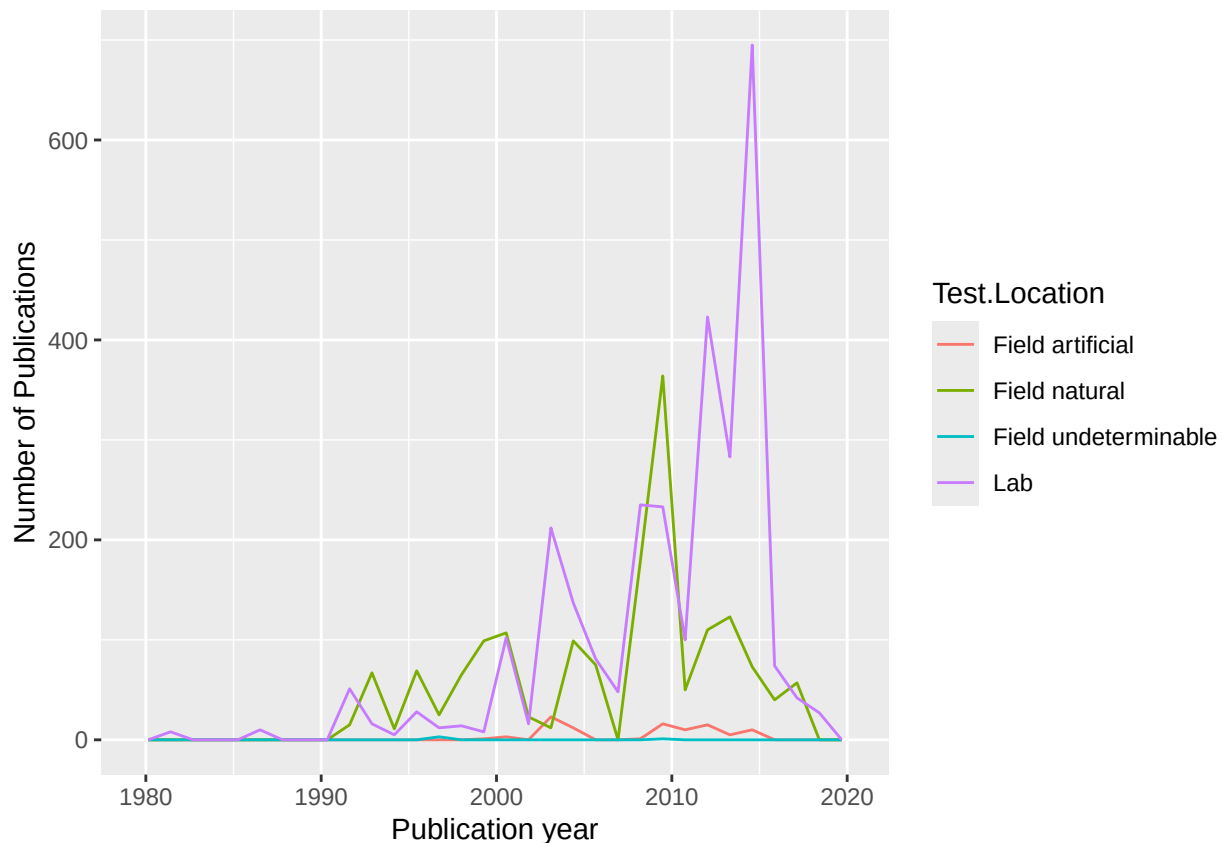
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics,aes(x=Publication.Year, color = Test.Location)) +  
  geom_freqpoly() +  
  xlab("Publication year") +  
  ylab("Number of Publications")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: Overall, the most common test locations are in the lab, but in the mid 1990s and late 2000s, natural field locations were more common. Field artificial have never really been that common, and recently field natural has become less common with lab becoming much more common.

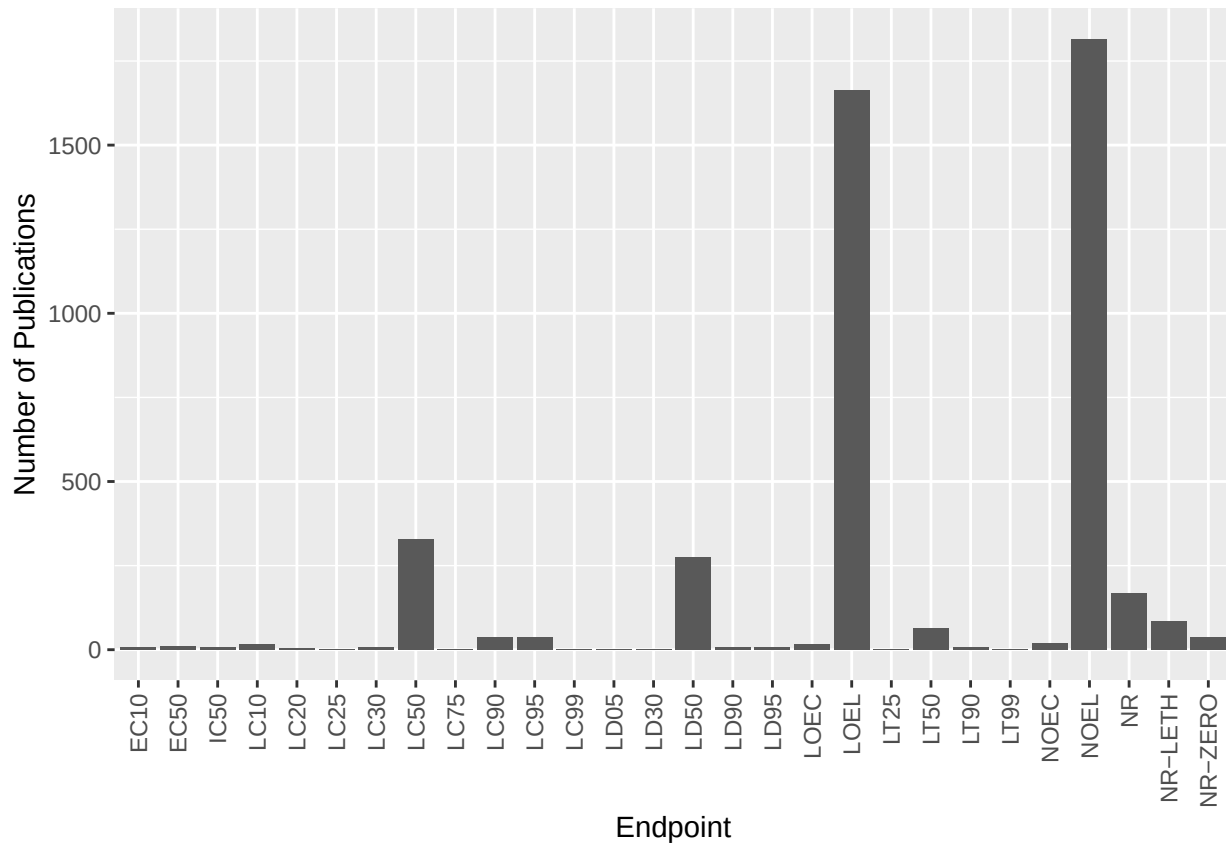
11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
Neonics$Endpoint <- as.factor(Neonics$Endpoint)
summary(Neonics$Endpoint)
```

##	EC10	EC50	IC50	LC10	LC20	LC25	LC30	LC50	LC75	LC90
##	6	11	6	15	5	1	6	327	1	37
##	LC95	LC99	LD05	LD30	LD50	LD90	LD95	LOEC	LOEL	LT25
##	36	2	1	1	274	6	7	17	1664	1
##	LT50	LT90	LT99	NOEC	NOEL	NR	NR-LETH	NR-ZERO		
##	65	7	2	19	1816	167	86	37		

```
ggplot(Neonics, aes(x=Endpoint)) +
  geom_bar() +
  xlab("Endpoint") +
  ylab("Number of Publications") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common endpoints are NOEL (no observable effect level) and LOEL (lowest observable effect level).

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
names(Litter)
```

```
## [1] "uid"                "namedLocation"
## [3] "domainID"          "siteID"
## [5] "plotID"            "trapID"
## [7] "weighDate"         "setDate"
## [9] "collectDate"       "ovenStartDate"
## [11] "ovenEndDate"       "fieldSampleID"
## [13] "massSampleID"      "samplingProtocolVersion"
## [15] "functionalGroup"   "dryMass"
## [17] "qaDryMass"         "remarks"
## [19] "measuredBy"
```

```
summary(Litter$collectDate)
```

```
## 2018-08-02 2018-08-30
##          91          97
```

```
Litter$collectDate <- ymd(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```



```
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

```
summary(Litter$plotID)
```

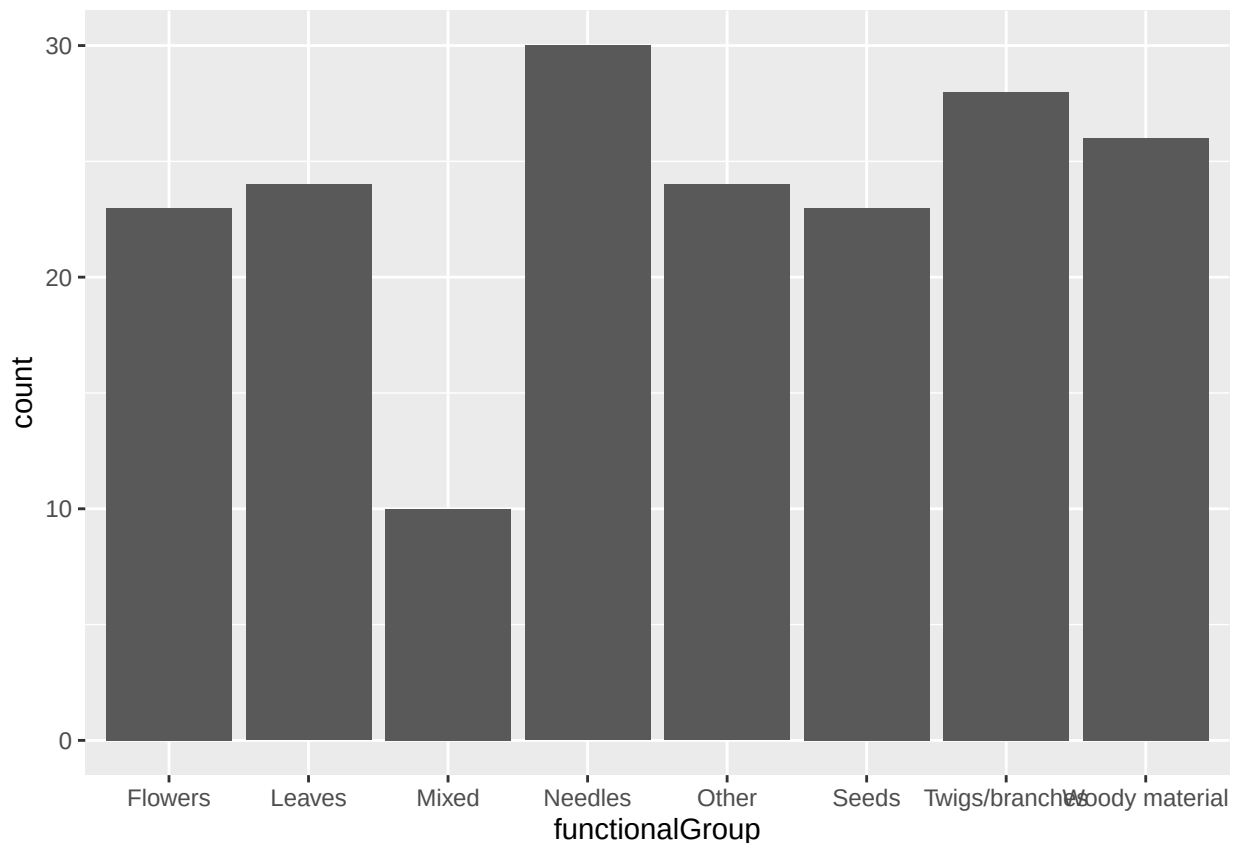
```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061  
##      20      19      18      15      14       8      16      17  
## NIWO_062 NIWO_063 NIWO_064 NIWO_067  
##      14      14      16      17
```

Answer: Summary outputs the class and mode, while Unique outputs the number of times each unique value occurs.

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

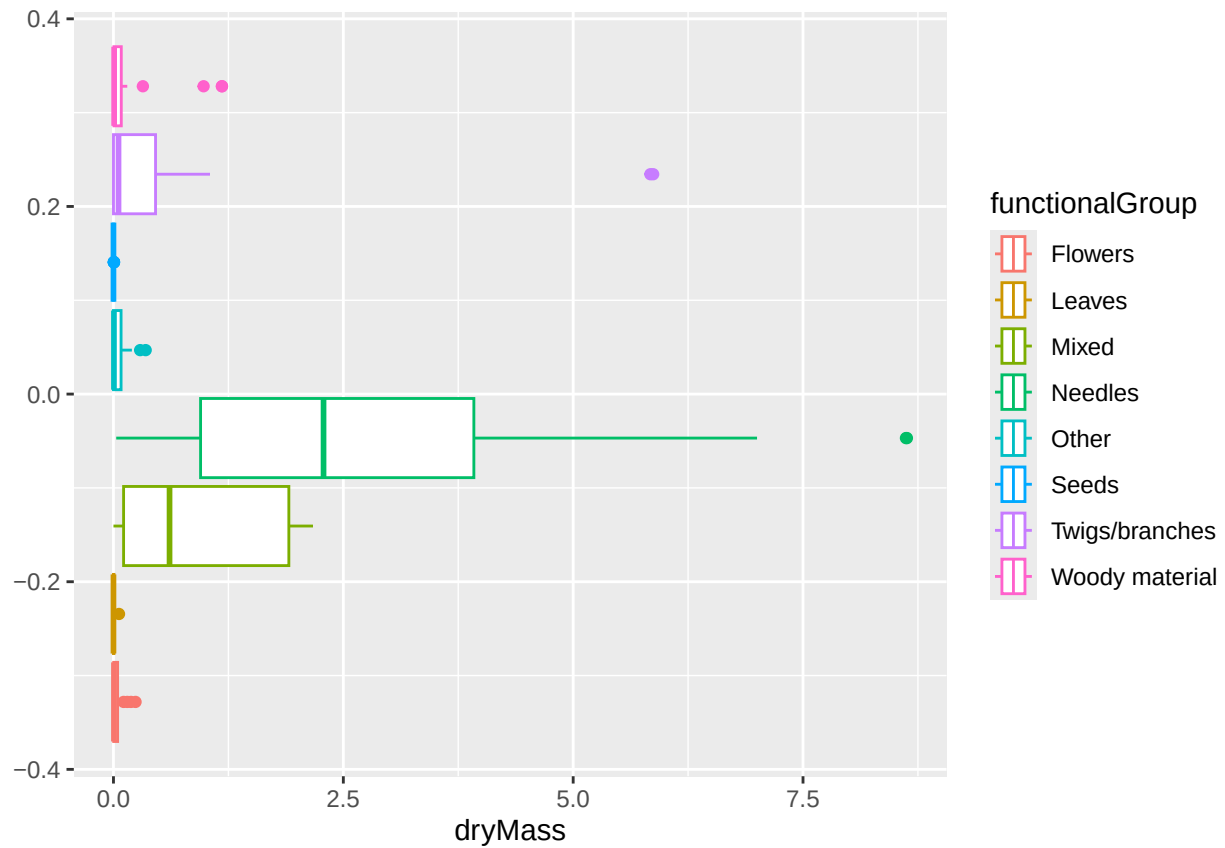
```
Litter$functionalGroup <- as.factor(Litter$functionalGroup)
```

```
ggplot(Litter, aes(x=functionalGroup)) +  
  geom_bar()
```

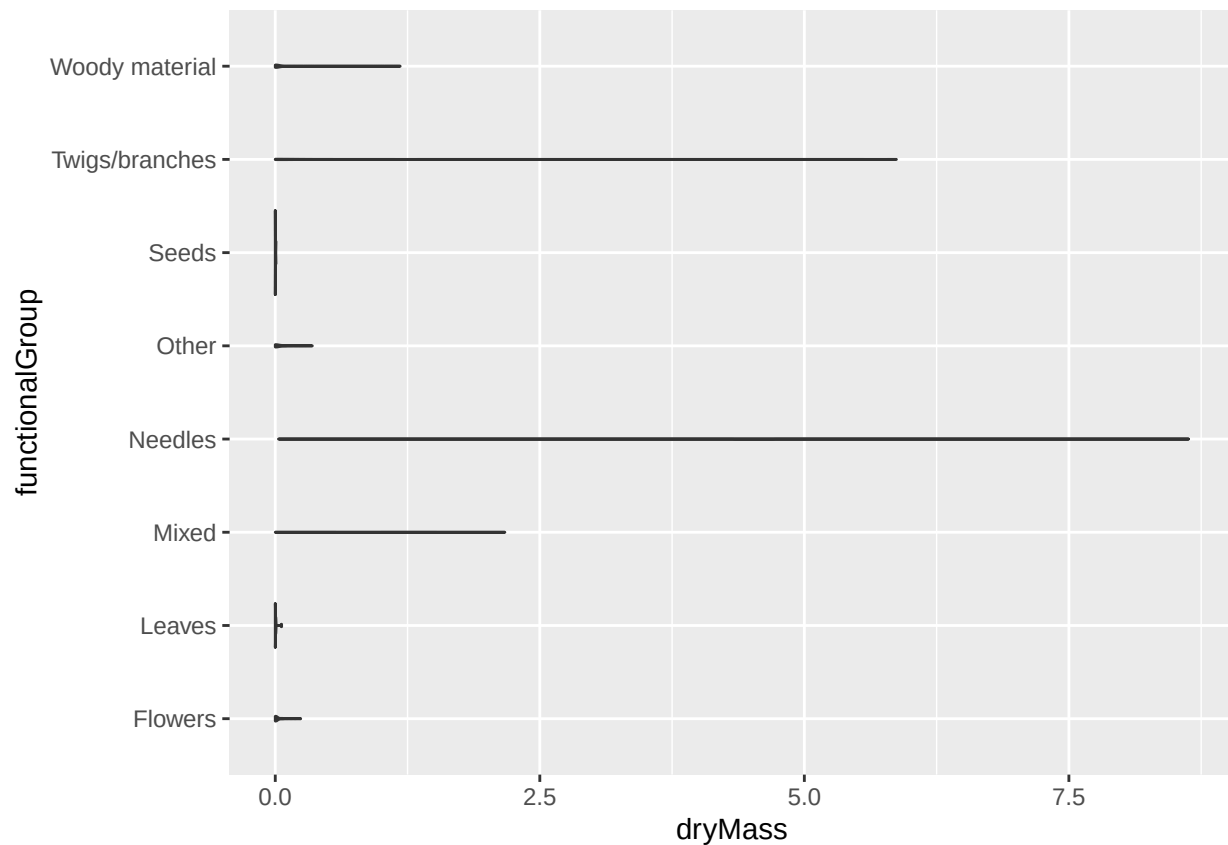


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter, aes(x=dryMass, color = functionalGroup)) +  
  geom_boxplot()
```



```
ggplot(Litter, aes(x=dryMass, y = functionalGroup)) +  
  geom_violin()
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: It shows the distribution of different types of litter to see which ones have the most biomass and is more clear on what it is showing.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles